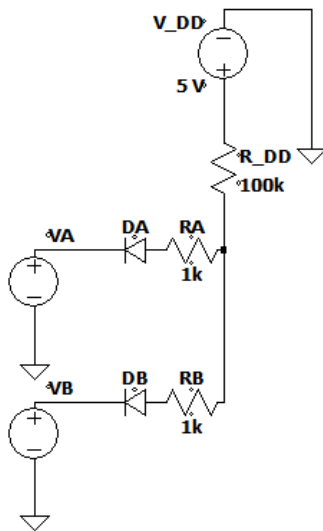


Basic Operation

Exercise 1



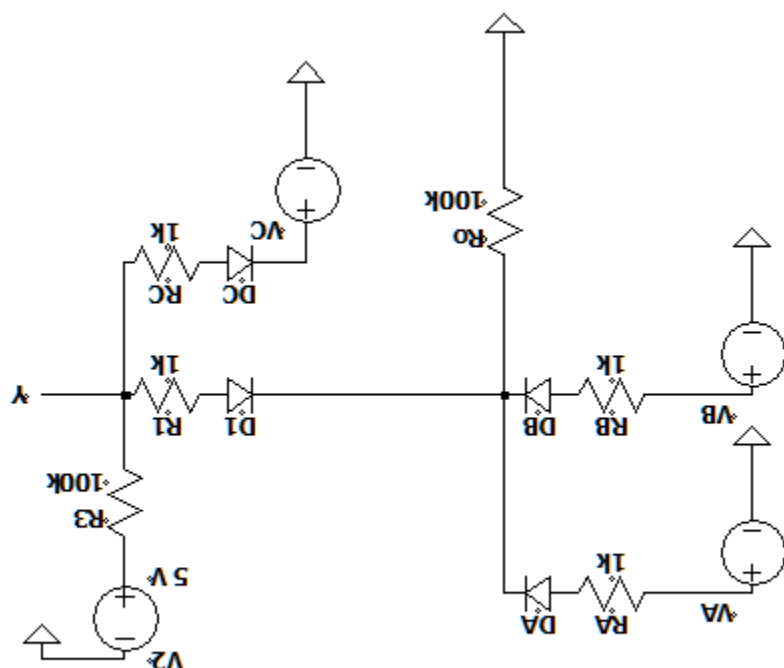
For the AND gate-

- For logic case (1,0), find the output voltage & verify your assumption of the diode model used.
- Find the current through R_{DD} for logic case (0,1).
- If R_B is doubled, find the currents through the diodes for logic case (0,0).
- Find the voltage of the node between D_B & R_B for the case in (c)

Ans: a) 0.743 V
b) 0.043 mA
c) 0.028 mA, 0.014 mA
d) 0.7 V

Practice Problem 1:

Implement the Boolean function, $Y = (A+B)C$ using diode logic.



Ans:

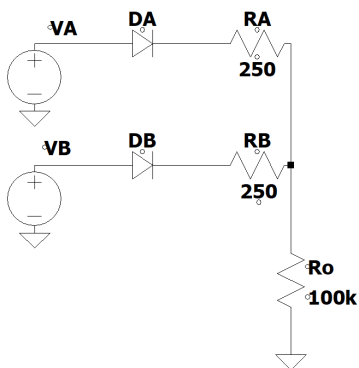
Exercise 2

Find the lower threshold voltage for the AND gate in **Exercise 1**, so that the output logic state can be determined correctly.

Ans: 0.743 V

Power Dissipation

Exercise 3



Find the power dissipation for all input logic cases.

[High input = 5 V, Low input = 0.2 V]

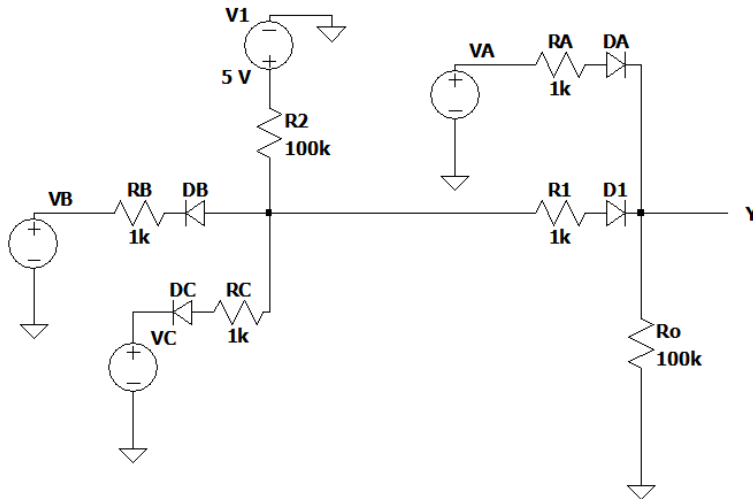
Ans: 0 mW, 2.145 μ W, 214.5 μ W, 214.5 μ W

Practice Problem 2:

Find the maximum & average power dissipation for the OR gate in **Exercise 3**.

Ans: 21.45 μ W, 16.0875 μ W

Exercise 4



For the diode logic circuit given-

- Find the Boolean expression of Y.
- Determine the higher & lower threshold of output voltage.
- Find the maximum & average power dissipation of the full circuit.

Ans:

a) $Y = A + BC$

b) 2.14 V, 0 V

c) 0.464 mW, 0.299 mW